Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Arnaldo P. & Maria E. Avila Dairy

Physical address of dairy:

8178 Paradise RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1966

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X065-X003-XXXX

B. OPERATORS

Avila, George

Operator name: Avila, George

Telephone no.: (209) 523-0256 Landline, (209) 988-0694 Cellular

8178 Paradise RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Avila, George

Legal owner name: Avila, George

Telephone no.: (209) 523-0256 Landline, (209) 988-0694 Cellular

8178 Paradise RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	200	85	150	150	60	0
Number under roof	300	0	0	0	0	0
Maximum number	500	85	150	150	60	0
Average number	500	85	150	150	60	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 17,406.65 *tons per reporting period*

Total nitrogen from manure: 216,273.01 *lbs per reporting period*

After ammonia losses (30% loss applied): 151,391.11 *lbs per reporting period*

Total phosphorus from manure: 35,753.52 *lbs per reporting period*

Total potassium from manure: 96,411.61 *lbs per reporting period*

Total salt from manure: 254,970.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 22,572,000 *gallons*

Total nitrogen generated: 35,848.09 *lbs*

Total phosphorus generated: 9,170.99 *lbs*

Total potassium generated: 40,976.92 *lbs*

Total salt generated: 425,070.89 *lbs*

	22,572,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	22,572,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 5	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	24	24	2	manure	X017-X065-X017-XXXX
Field 2	53	53	2	both	X017-X065-X003-XXXX
Field 3	35	35	2	both	X017-X065-X016-XXXX
Field 4	63	63	2	both	X017-X065-X002-XXXX
Field 5	27	27	2	both	X017-X065-X002-XXXX
Field 6	30	30	1	manure	X017-X065-X003-XXXX
Field 7	24	24	1	process wastewater	X017-X065-X003-XXXX
Field 8	33	33	1	manure	X017-X065-X003-XXXX
Totals for areas that were used for application	289	289	13		
Totals for areas that were not used for application					
Land application area totals	289	289	13		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	194.50 *ton*	As-is		57.3	9,600.00	1,500.00	9,400.00		8.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.10	155.60	24.31	152.36	598.66

Field 1

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	854.18 *ton*	As-is		70.0	3,000.00	800.00	3,600.00		5.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	35.59	213.55	56.95	256.25	1,165.96

Field 2

Field name: Field 2

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	972.22 *ton*	As-is		71.4	3,600.00	1,000.00	4,800.00		8.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.34	132.08	36.69	176.10	938.04

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	1,387.77 *ton*	As-is		65.8	3,800.00	700.00	2,600.00		4.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.18	199.00	36.66	136.16	757.60

Field 3

Field name: Field 3

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	659.50 *ton*	As-is		70.5	3,400.00	900.00	55,300.00		8.22

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.84	128.13	33.92	2,084.02	913.84

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,214.94 *ton*	As-is		66.8	4,200.00	1,000.00	4,000.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	34.71	291.59	69.43	277.70	1,267.70

Field 4

Field name: Field 4

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,221.50 *ton*	As-is		70.7	3,100.00	900.00	4,100.00		7.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.39	120.21	34.90	158.99	902.13

Field 4

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	1,649.62 *ton*	As-is		64.8	3,700.00	800.00	3,600.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.18	193.76	41.90	188.53	811.09

Field 5

Field name: Field 5

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	461.70 *ton*	As-is		69.3	4,600.00	1,200.00	7,600.00		9.69

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.10	157.32	41.04	259.92	1,017.39

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	706.98 *ton*	As-is		63.6	4,200.00	900.00	4,700.00		4.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.18	219.95	47.13	246.13	924.52

Field 6

Field name: Field 6

11/17/2022: Oats, hay

Crop: Oats, hay Acres planted: 30 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 7

Field name: Field 7

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 8

Field name: Field 8

Field 8

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/10/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	233.05	109.56	490.01	8,656.87	410.00 *ton*
Application event totals		233.05	109.56	490.01	8,656.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.47	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.47	

Field 1 - 05/27/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	264.96	108.93	400.38	7,418.88	480.00 *ton*
Application event totals		264.96	108.93	400.38	7,418.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	44.29	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	44.29	

Field 1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.08	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	39.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.08	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	39.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.47	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	33.87	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	33.87	

Field 2 - 11/19/2022: Oats, silage-soft dough

Field 2 - 11/19/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.37	10.12	53.93	0.00	855,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	79.07	6,277,500.00 *gal*
Application event totals		45.37	10.12	53.93	79.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	11.63	2.68	14.86	138.57	456,000.00 *gal*
Application event totals		11.63	2.68	14.86	138.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.54	3.35	18.58	173.21	570,000.00 *gal*
Application event totals		14.54	3.35	18.58	173.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.54	3.35	18.58	173.21	570,000.00 *gal*
Application event totals		14.54	3.35	18.58	173.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.45	4.02	22.29	207.86	684,000.00 *gal*
Application event totals		17.45	4.02	22.29	207.86	

Field 2 - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.57	3.89	22.05	297.51	741,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	30.67	5,265,000.00 *gal*
Application event totals		16.57	3.89	22.05	328.19	

04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.93	3.74	21.20	286.07	712,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	29.49	5,062,500.00 *gal*
Application event totals		15.93	3.74	21.20	315.56	

Field 2 - 06/28/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	102.04	32.55	66.36	3,987.62	530.00 *ton*
Application event totals		102.04	32.55	66.36	3,987.62	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.12	4.48	25.44	343.29	855,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.39	6,075,000.00 *gal*
Application event totals		19.12	4.48	25.44	378.68	

Field 2 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.38	7.05	26.42	339.65	826,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.21	5,872,500.00 *gal*
Application event totals		25.38	7.05	26.42	373.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.38	7.05	26.42	339.65	826,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.21	5,872,500.00 *gal*
Application event totals		25.38	7.05	26.42	373.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.38	7.05	26.42	339.65	826,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.21	5,872,500.00 *gal*
Application event totals		25.38	7.05	26.42	373.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.50	6.81	25.51	327.94	798,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	33.03	5,670,000.00 *gal*
Application event totals		24.50	6.81	25.51	360.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.75	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.75	

Field 2 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.39	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.39	

Field 3 - 11/10/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	233.87	109.94	491.72	8,687.03	600.00 *ton*
Application event totals		233.87	109.94	491.72	8,687.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	5.79	1.36	7.71	103.97	171,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		5.79	1.36	7.71	139.70	

Field 3 - 05/27/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	158.98	65.36	240.23	4,451.33	420.00 *ton*
Application event totals		158.98	65.36	240.23	4,451.33	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.26	4.75	26.97	363.88	598,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	37.51	4,252,500.00 *gal*
Application event totals		20.26	4.75	26.97	401.40	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.30	4.53	25.69	346.56	570,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		19.30	4.53	25.69	382.28	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.50	7.37	27.59	354.71	570,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		26.50	7.37	27.59	390.44	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.50	7.37	27.59	354.71	570,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		26.50	7.37	27.59	390.44	

Field 3 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.30	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.30	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.18	7.00	26.21	336.97	541,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	33.94	3,847,500.00 *gal*
Application event totals		25.18	7.00	26.21	370.92	

Field 4 - 11/18/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	174.56	28.31	28.98	3,890.82	1,100.00 *ton*
Application event totals		174.56	28.31	28.98	3,890.82	

Field 4 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	5.36	1.26	7.13	96.27	285,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	31.76	6,480,000.00 *gal*
Application event totals		5.36	1.26	7.13	128.02	

Field 4 - 06/29/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	102.04	32.55	66.36	3,987.62	630.00 *ton*
Application event totals		102.04	32.55	66.36	3,987.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Field 4 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	997,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	7,087,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	997,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	7,087,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	997,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	7,087,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	997,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	7,087,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.03	6.96	26.06	335.00	969,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	33.74	6,885,000.00 *gal*
Application event totals		25.03	6.96	26.06	368.75	

Field 4 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	7,087,500.00 *gal*
Application event totals		0.00	0.00	0.00	34.74	

Field 5 - 11/18/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.55	12.59	67.05	0.00	541,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	70.10	2,835,000.00 *gal*
Application event totals		56.55	12.59	67.05	70.10	

01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.96	9.20	51.05	476.02	798,000.00 *gal*
Application event totals		39.96	9.20	51.05	476.02	

01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.66	10.51	58.35	544.02	912,000.00 *gal*
Application event totals		45.66	10.51	58.35	544.02	

Field 5 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.26	3.81	21.64	292.01	370,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	30.10	2,632,500.00 *gal*
Application event totals		16.26	3.81	21.64	322.11	

04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.01	3.52	19.98	269.54	342,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	27.79	2,430,000.00 *gal*
Application event totals		15.01	3.52	19.98	297.33	

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

06/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	123.55	100.59	280.99	7,926.67	300.00 *ton*
Application event totals		123.55	100.59	280.99	7,926.67	

Field 5 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	34.74	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	427,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	3,037,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	427,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	3,037,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	427,500.00 *gal*
Application event totals		25.77	7.16	26.82	344.86	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.77	7.16	26.82	344.86	427,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	3,037,500.00 *gal*
Application event totals		25.77	7.16	26.82	379.59	

Field 5 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.05	6.68	25.03	321.87	399,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	32.42	2,835,000.00 *gal*
Application event totals		24.05	6.68	25.03	354.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.05	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.05	

Field 6 - 11/17/2022: Oats, hay

Field name: Field 6

Crop: Oats, hay Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	169.96	27.56	28.22	3,788.24	510.00 *ton*
Application event totals		169.96	27.56	28.22	3,788.24	

Field 7 - 11/17/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.27	13.41	71.46	0.00	513,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	67.59	2,430,000.00 *gal*
Application event totals		60.27	13.41	71.46	67.59	

Field 8 - 11/17/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

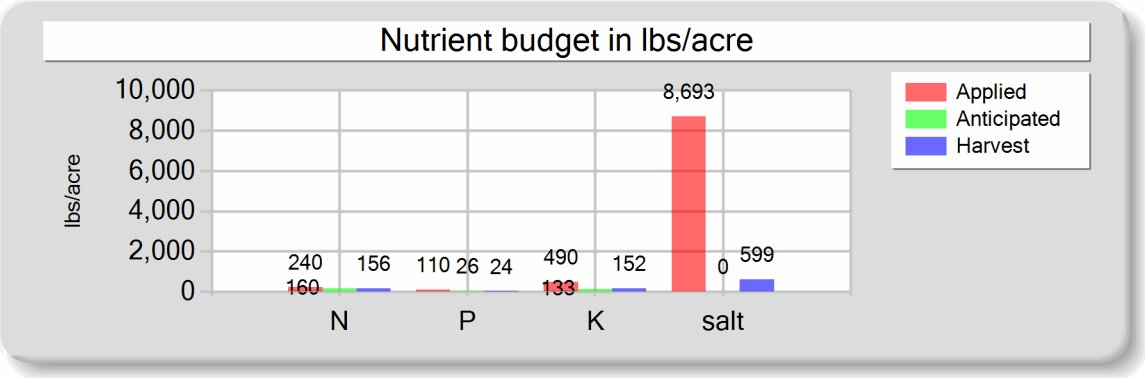
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	178.74	28.99	29.68	3,984.07	590.00 *ton*
Application event totals		178.74	28.99	29.68	3,984.07	

B. NUTRIENT BUDGET

Field 1 - 11/10/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.05	109.56	490.01	8,656.87
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	36.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.05	109.56	490.01	8,693.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	155.60	24.31	152.36	598.66
Nutrient balance	84.45	85.24	337.65	8,094.68
Applied to removed ratio	1.54	4.51	3.22	14.52

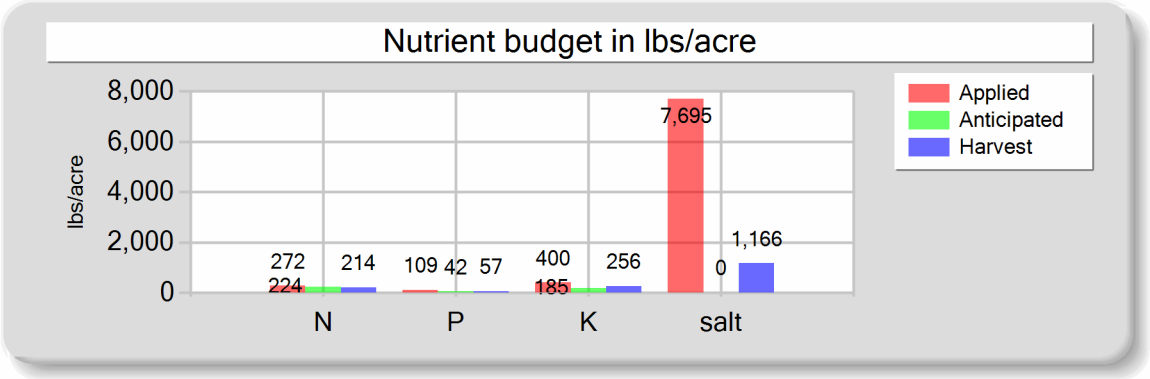
Fresh water applied	
2,835,000.00	*gallons*
104.40	*acre-inches*
4.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/27/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	264.96	108.93	400.38	7,418.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	276.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.96	108.93	400.38	7,695.03
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	213.55	56.95	256.25	1,165.96
Nutrient balance	58.42	51.98	144.13	6,529.08
Applied to removed ratio	1.27	1.91	1.56	6.60

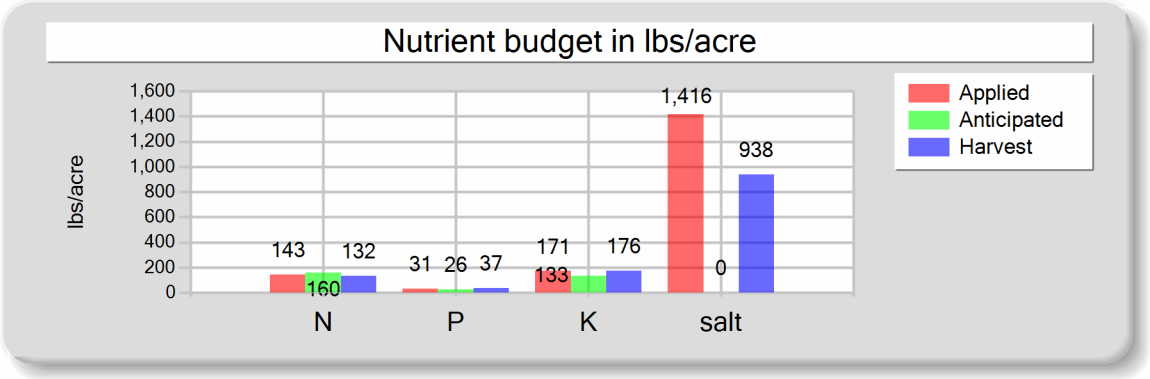
Fresh water applied	
21,465,000.00	*gallons*
790.48	*acre-inches*
32.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/19/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	136.02	31.13	171.50	1,276.44
Fresh water	0.00	0.00	0.00	139.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	143.02	31.13	171.50	1,415.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.08	36.69	176.10	938.04
Nutrient balance	10.95	-5.55	-4.61	477.64
Applied to removed ratio	1.08	0.85	0.97	1.51

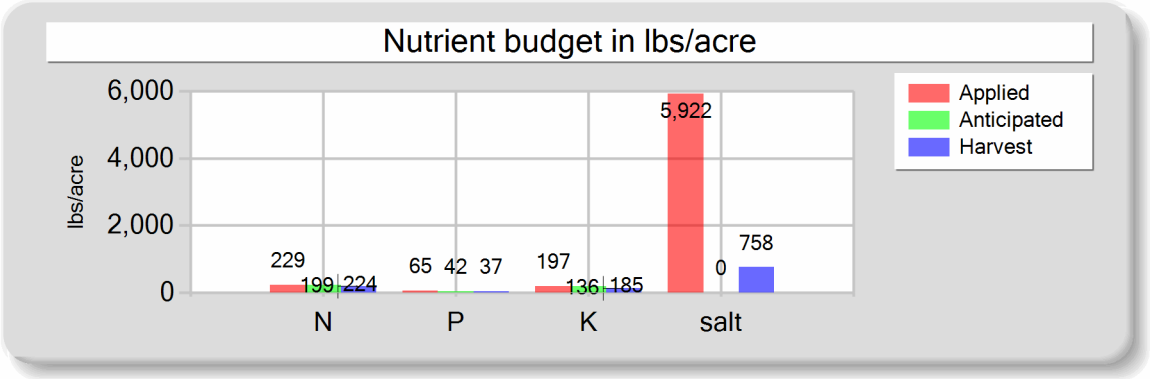
Fresh water applied	
16,605,000.00	*gallons*
611.51	*acre-inches*
11.54	*inches/acre*

Process wastewater applied	
4,588,500.00	*gallons*
168.98	*acre-inches*
3.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/28/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	102.04	32.55	66.36	3,987.62
Process wastewater	119.75	32.45	130.20	1,690.18
Fresh water	0.00	0.00	0.00	244.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.78	65.00	196.56	5,922.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	199.00	36.66	136.16	757.60
Nutrient balance	29.78	28.35	60.40	5,164.41
Applied to removed ratio	1.15	1.77	1.44	7.82

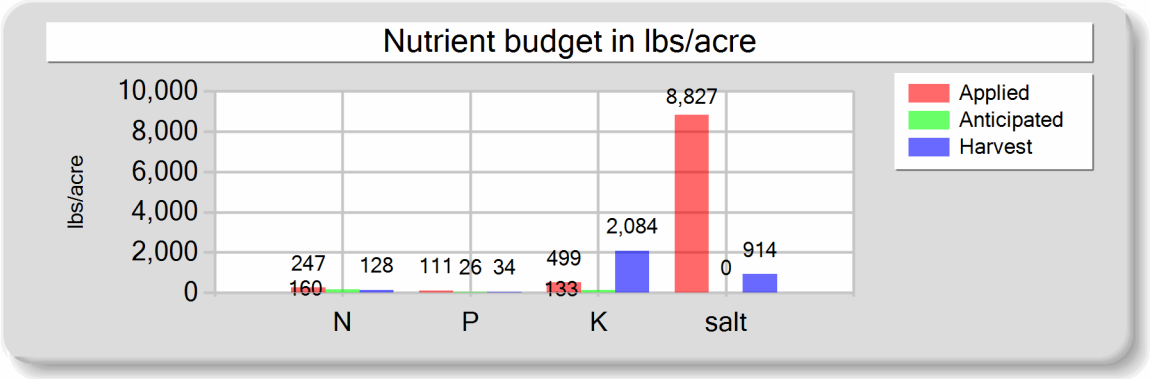
Fresh water applied	
41,917,500.00	*gallons*
1,543.68	*acre-inches*
29.13	*inches/acre*

Process wastewater applied	
4,132,500.00	*gallons*
152.19	*acre-inches*
2.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/10/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.87	109.94	491.72	8,687.03
Process wastewater	5.79	1.36	7.71	103.97
Fresh water	0.00	0.00	0.00	35.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.66	111.29	499.42	8,826.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	128.13	33.92	2,084.02	913.84
Nutrient balance	118.52	77.38	-1,584.60	7,912.89
Applied to removed ratio	1.93	3.28	0.24	9.66

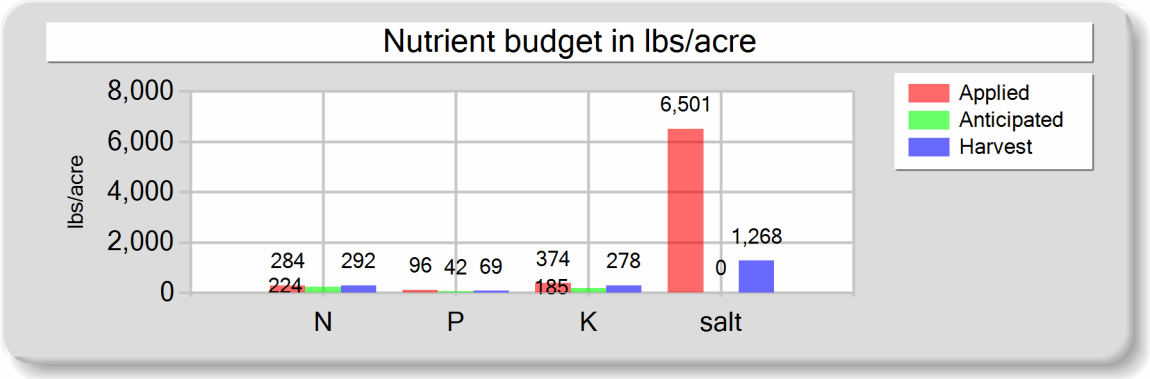
Fresh water applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
4.26	*inches/acre*

Process wastewater applied	
171,000.00	*gallons*
6.30	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/27/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.98	65.36	240.23	4,451.33
Process wastewater	117.74	31.01	134.04	1,756.83
Fresh water	0.00	0.00	0.00	292.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.72	96.36	374.27	6,501.14
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	291.59	69.43	277.70	1,267.70
Nutrient balance	-7.87	26.94	96.57	5,233.43
Applied to removed ratio	0.97	1.39	1.35	5.13

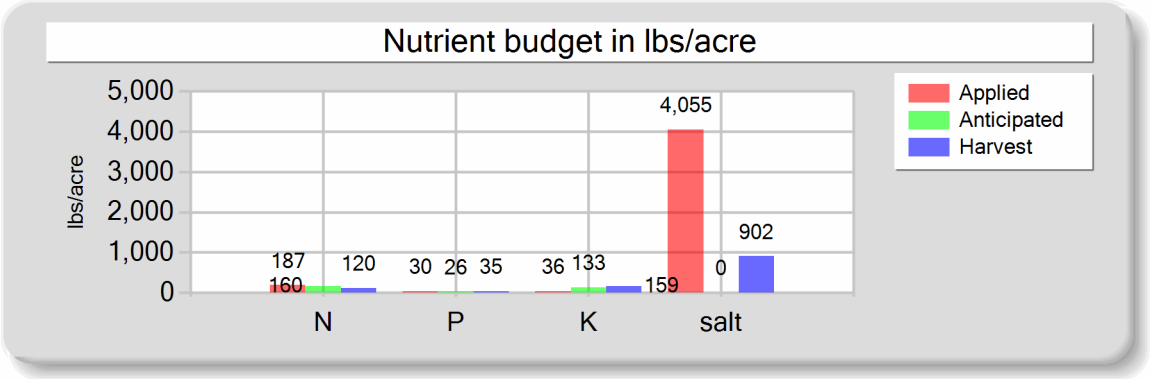
Fresh water applied	
33,210,000.00	*gallons*
1,223.01	*acre-inches*
34.94	*inches/acre*

Process wastewater applied	
2,850,000.00	*gallons*
104.96	*acre-inches*
3.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/18/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	174.56	28.31	28.98	3,890.82
Process wastewater	5.36	1.26	7.13	96.27
Fresh water	0.00	0.00	0.00	67.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	186.92	29.56	36.12	4,054.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	120.21	34.90	158.99	902.13
Nutrient balance	66.71	-5.34	-122.87	3,152.44
Applied to removed ratio	1.55	0.85	0.23	4.49

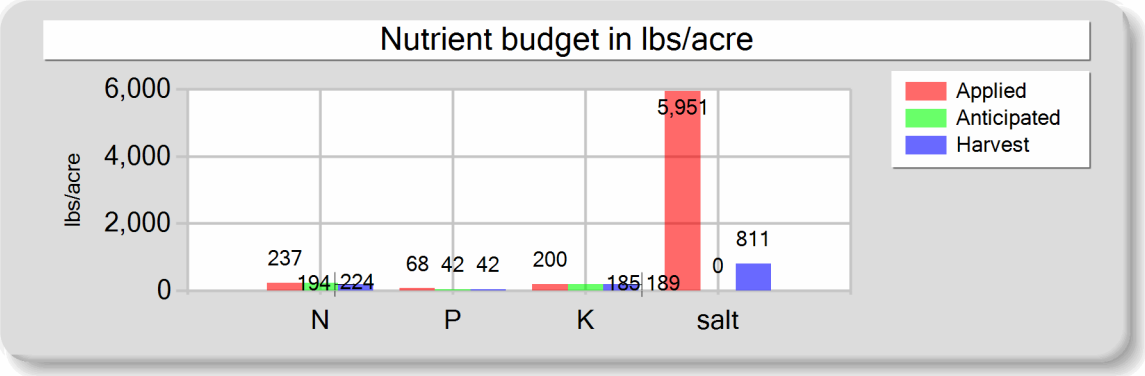
Fresh water applied	
13,770,000.00	*gallons*
507.10	*acre-inches*
8.05	*inches/acre*

Process wastewater applied	
285,000.00	*gallons*
10.50	*acre-inches*
0.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/29/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	102.04	32.55	66.36	3,987.62
Process wastewater	128.09	35.60	133.34	1,714.43
Fresh water	0.00	0.00	0.00	249.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.13	68.15	199.70	5,951.16
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	193.76	41.90	188.53	811.09
Nutrient balance	43.36	26.26	11.17	5,140.07
Applied to removed ratio	1.22	1.63	1.06	7.34

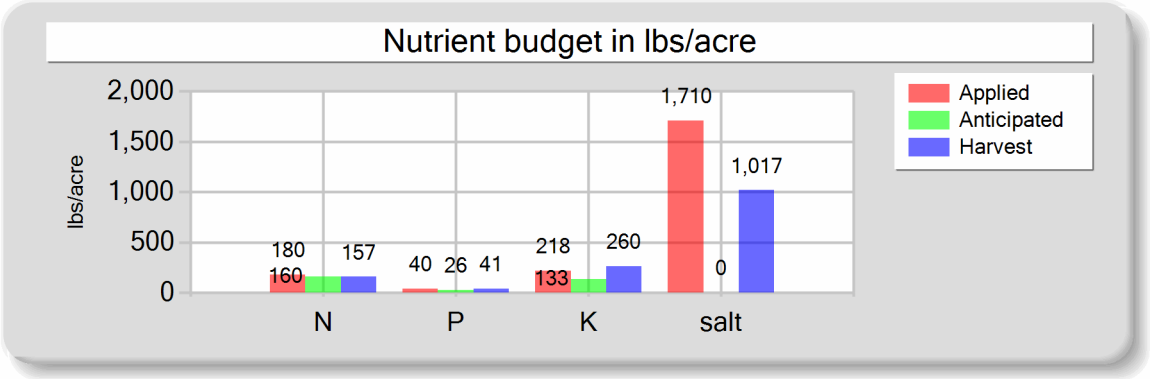
Fresh water applied	
50,827,500.00	*gallons*
1,871.80	*acre-inches*
29.71	*inches/acre*

Process wastewater applied	
4,959,000.00	*gallons*
182.62	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/18/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.44	39.63	218.07	1,581.59
Fresh water	0.00	0.00	0.00	127.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.44	39.63	218.07	1,709.58
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.32	41.04	259.92	1,017.39
Nutrient balance	23.12	-1.41	-41.85	692.19
Applied to removed ratio	1.15	0.97	0.84	1.68

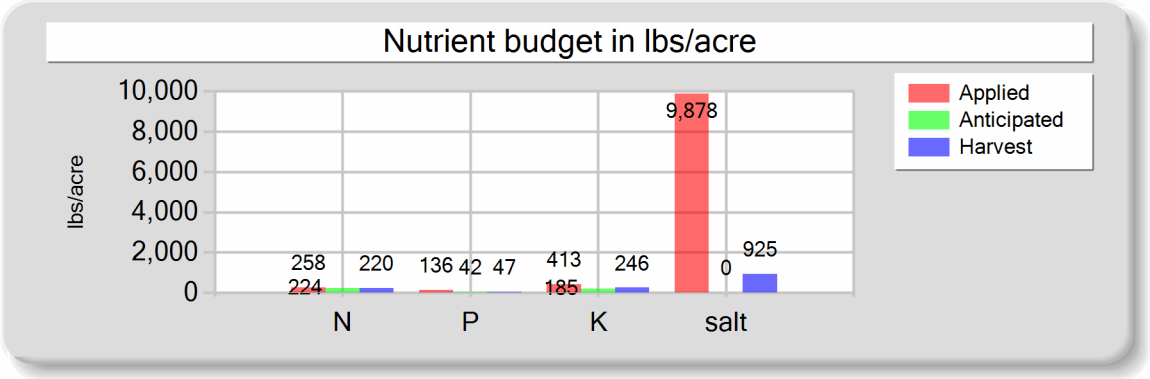
Fresh water applied	
7,897,500.00	*gallons*
290.84	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
2,964,000.00	*gallons*
109.15	*acre-inches*
4.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123.55	100.59	280.99	7,926.67
Process wastewater	127.11	35.33	132.32	1,701.30
Fresh water	0.00	0.00	0.00	250.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.65	135.92	413.31	9,878.06
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	219.95	47.13	246.13	924.52
Nutrient balance	37.71	88.78	167.18	8,953.54
Applied to removed ratio	1.17	2.88	1.68	10.68

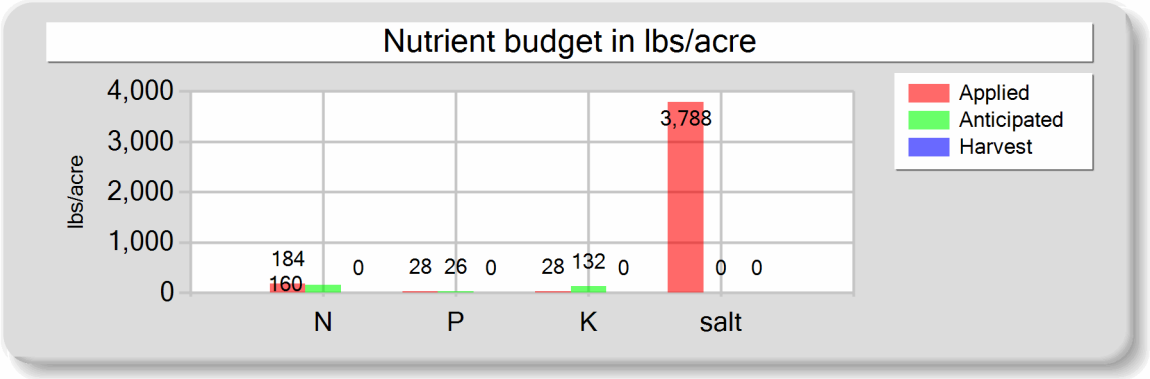
Fresh water applied	
21,870,000.00	*gallons*
805.40	*acre-inches*
29.83	*inches/acre*

Process wastewater applied	
2,109,000.00	*gallons*
77.67	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/17/2022: Oats, hay

Field name: Field 6 Crop: Oats, hay Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.96	27.56	28.22	3,788.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	183.96	27.56	28.22	3,788.24
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	183.96	27.56	28.22	3,788.24
Applied to removed ratio	0.00	0.00	0.00	0.00

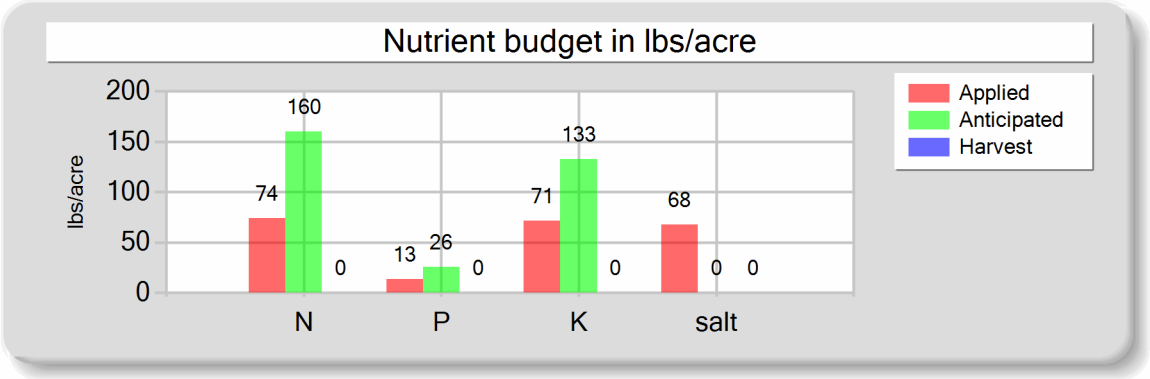
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 7 - 11/17/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	60.27	13.41	71.46	0.00
Fresh water	0.00	0.00	0.00	67.59
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	74.27	13.41	71.46	67.59
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	74.27	13.41	71.46	67.59
Applied to removed ratio	0.00	0.00	0.00	0.00

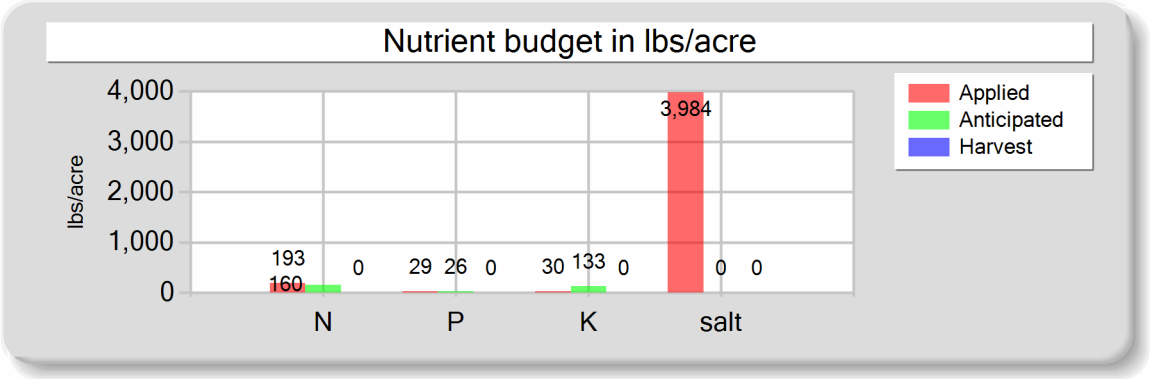
Fresh water applied	
2,430,000.00	*gallons*
89.49	*acre-inches*
3.73	*inches/acre*

Process wastewater applied	
513,000.00	*gallons*
18.89	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/17/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.74	28.99	29.68	3,984.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	192.74	28.99	29.68	3,984.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	192.74	28.99	29.68	3,984.07
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Cow Corral

Sample and source description: 2022 Cow Corral

Sample date: 11/04/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	5,500.00	24,600.00	15,100.00	8,000.00	8,800.00	3,400.00	1,450.00		43.46
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

2022 Settling Basin

Sample and source description: 2022 Settling Basin

Sample date: 11/04/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	4,200.00	4,300.00	15,200.00	5,400.00	1,900.00	4,800.00	140.00		57.73
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Heifer Corral

Sample and source description: Heifer Corral

Sample date: 04/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,000.00	7,400.00	27,200.00	10,900.00	3,200.00	0.54	3,000.00	20,300.00		50.40
DL	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.02		0.10

Dry Cow Corral

Sample and source description: Dry Cow Corral

Sample date: 06/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	9,200.00	25,700.00	14,600.00	4,600.00	4,500.00	2,800.00	10,500.00		72.50
DL	0.01	0.20	0.02	0.02	0.02	0.02	0.02	0.02		0.10

Settling Lagoon

Sample and source description: Settling Lagoon

Sample date: 06/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	5,200.00	10,600.00	13,800.00	4,600.00	3,800.00	2,700.00	5,500.00		63.70
DL	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.02		0.10

Settling Pond

Sample and source description: Settling Pond

Sample date: 12/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	3,400.00	2,600.00	10,600.00	2,400.00	1,100.00	2,400.00	1,300.00		53.80
DL	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.02		0.10

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 03/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	162.00	132.00	0.00	0.00	37.30	207.00								2,910.00	1,930
DL	10.00	2.00	0.50	0.50	0.20	0.50								0.10	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	142.00	129.00	0.00	0.00	33.30	189.00								3,840.00	2,550
DL	10.00	2.00	0.50	0.50	0.20	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 10/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	195.00	155.00	0.00	0.00	54.20	203.00								3,930.00	2,610
DL	10.00	2.00	0.50	0.50	0.20	0.50								0.10	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	319.00	231.00	0.00	0.00	67.70	346.00								5,130.00	3,400
DL	10.00	2.00	0.50	0.50	0.20	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 5

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	80
DL	0.50	0.50	0.50								1.00	10

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								38.10	37
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			89.00				
DL			4.00				

Field 2

Field 2

Soils

Sample and source description: Soils

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			143.00				
DL			4.00				

Field 3

Soils

Sample and source description: Soils

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			250.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			62.00				
DL			4.00				

Field 5

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			91.00				
DL			4.00				

Field 7

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			73.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/10/2022: Oats, silage-soft dough

Field 1 - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	1,500.00	9,400.00		8.65
DL	0.05	0.02	0.02		0.05

Field 1 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	800.00	3,600.00		5.46
DL	0.05	0.02	0.02		0.05

Field 2 - 11/19/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	1,000.00	4,800.00		8.94
DL	0.05	0.02	0.02		0.05

Field 2 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	2,600.00		4.23
DL	0.05	0.02	0.02		0.05

Field 3 - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	900.00	55,300.00		8.22
DL	0.05	0.02	0.02		0.05

Field 3 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,000.00	4,000.00		5.50
DL	0.05	0.02	0.02		0.05

Field 4 - 11/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	900.00	4,100.00		7.94
DL	0.05	0.02	0.02		0.05

Field 4 - 06/29/2023: Corn, silage

Corn Slilage

Sample and source description: Corn Slilage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	800.00	3,600.00		4.40
DL	0.05	0.02	0.02		0.05

Field 5 - 11/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	1,200.00	7,600.00		9.69
DL	0.05	0.02	0.02		0.05

Field 5 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	900.00	4,700.00		4.85
DL	0.05	0.02	0.02		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

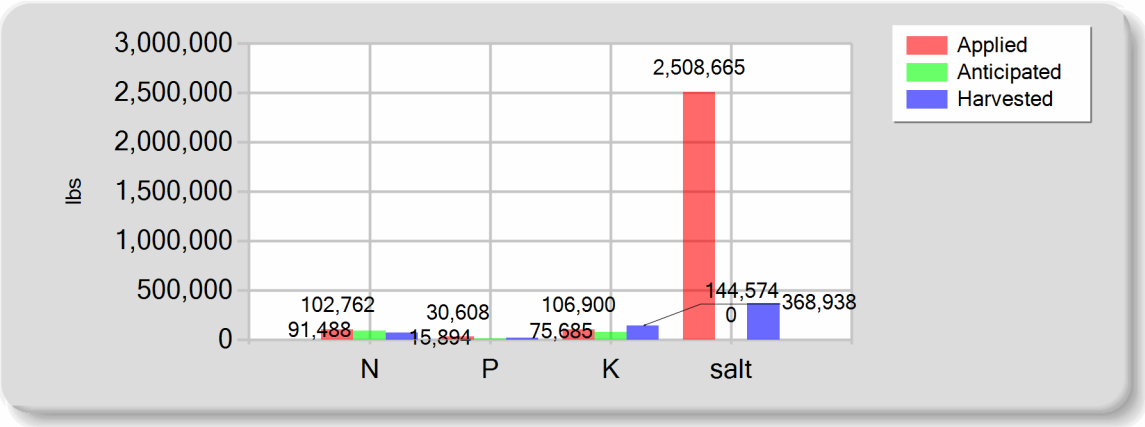
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

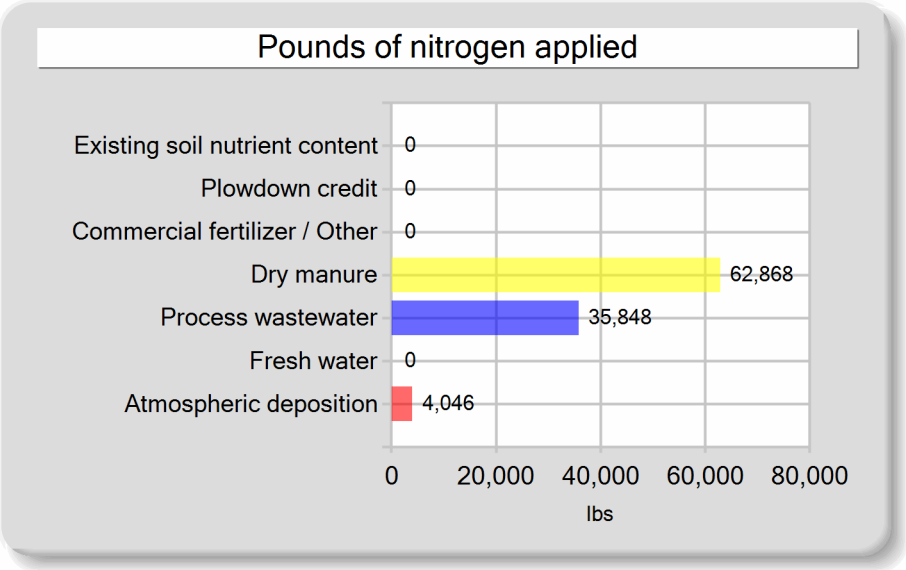
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

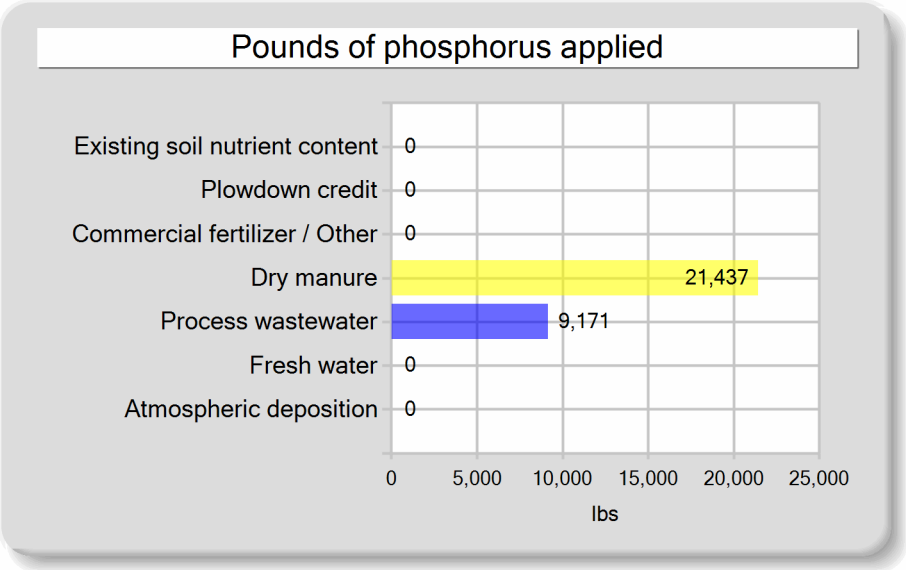
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	62,868.27	21,437.40	65,923.21	2,012,487.72
Process wastewater	35,848.09	9,170.99	40,976.92	425,070.89
Fresh water	0.00	0.00	0.00	71,106.03
Atmospheric deposition	4,046.00	0.00	0.00	0.00
Total nutrients applied	102,762.36	30,608.39	106,900.13	2,508,664.64
Anticipated crop nutrient removal	91,488.00	15,894.40	75,684.80	0.00
Actual crop nutrient removal	71,063.37	16,673.22	144,573.65	368,938.43
Nutrient balance	31,698.99	13,935.17	-37,673.51	2,139,726.21
Applied to removed ratio	1.45	1.84	0.74	6.80

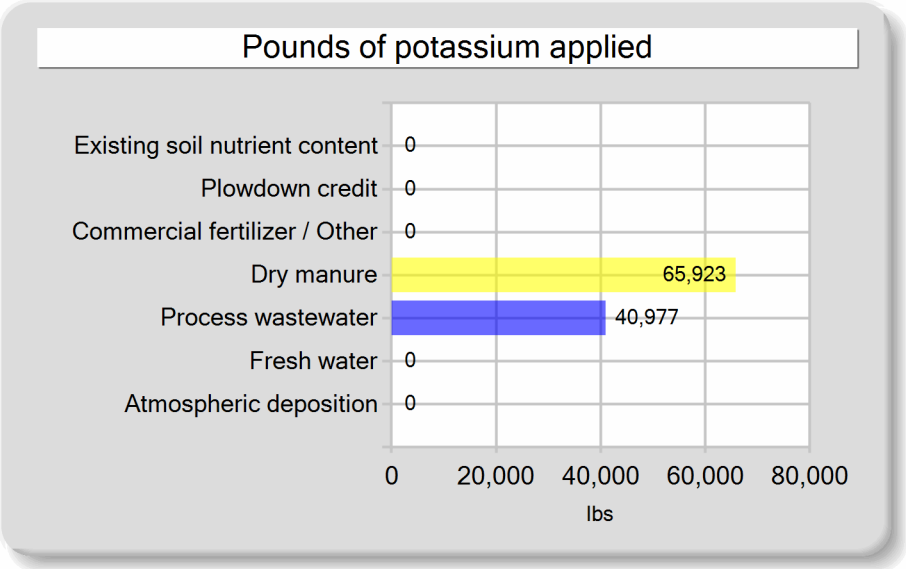
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

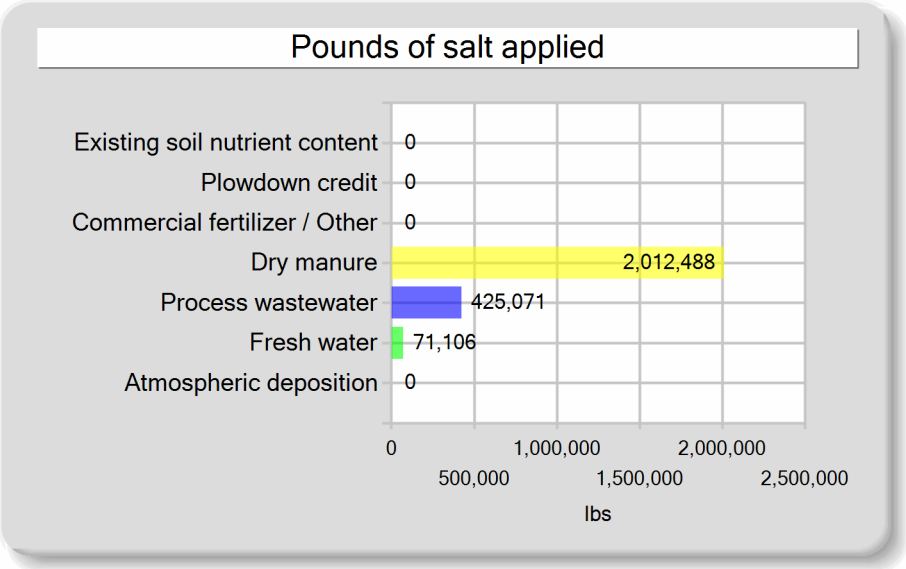


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 1 Forage (1.54), Field 3 Forage (1.93) and Field 4 Forage (1.55) all had high nitrogen field ratios as all 3 fields failed to meet anticipated nitrogen removal rate of 160 pounds/acre. The subject fields removed 155, 128 and 120 pounds of nitrogen/acre respectively. Fields 1 and 3 had a slight over application of nitrogen in the form of solid manure which was a factor in the high field ratios. Field 4 nitrogen application was not considered excessive but in-line with that recommended in the dairy's nutrient management plan. Yield for Field 4 was adequate however a low nitrogen value from the plant tissue analysis resulted in only 6.2 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton which ultimately resulted in the high nitrogen field ratio.

Fields 6, 7, and 8 (totaling 87 acres) located along the river were flooded out due to high flows. The subject fields were planted in oats in the fall of 2022 and were flooded in February of 2023, prior to harvest. Fields remained under water into the early summer and were not able to be planted in corn.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

George Avila
George Avila (May 5, 2024 09:51 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
George Avila	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 5, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
A.P. Avila & Sons
8178 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 10:00 AM
SUBMITTED BY: George
ANALYZED DATE: 6/7/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #1	T3158045A	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	684	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)	2.37
Electrical Conductivity (EC)	96.1			0.164
Nitrate Nitrogen (NO3-N)	105	112 (T3158049A)	109 (T3158049A)	2.08

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
A.P. Avila & Sons
8178 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 10:00 AM
SUBMITTED BY: George
ANALYZED DATE: 6/9/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #2	T3158045B	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	687	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)	2.37
Electrical Conductivity (EC)	96.1			0.164
Nitrate Nitrogen (NO3-N)	106	94.2 (W3158004A)	92.0 (W3158004A)	2.43

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
A.P. Avila & Sons
8178 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 10:00 AM
SUBMITTED BY: George
ANALYZED DATE: 6/9/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #3	T3158045C	Electrical Conductivity (EC)	EPA 120.1 and	0.920	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.60	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	589	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)	2.37
Electrical Conductivity (EC)	96.1			0.164
Nitrate Nitrogen (NO3-N)	106	94.2 (W3158004A)	92.0 (W3158004A)	2.43

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
A.P. Avila & Sons
8178 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 6/7/23 10:00 AM
SUBMITTED BY: George
ANALYZED DATE: 6/9/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #4	T3158045D	Electrical Conductivity (EC)	EPA 120.1 and	0.992	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	27.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	635	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	107	100 (T3158045A)	102 (T3158045A)	2.37
Electrical Conductivity (EC)	96.1			0.164
Nitrate Nitrogen (NO3-N)	106	94.2 (W3158004A)	92.0 (W3158004A)	2.43

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: A.P. Avila ~~Family Dairy~~
Mailing address: 8178 PARADISE Rd
(not facility address)
Phone#: (209) 526-1393 c(209) 988-0694

Authorized Copy Released To:

Sampled By: GEORGE Avila
Project ID: Domestic wells

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	6-7-23	9:15	well # 1		19.0°		✓				
2	" "	9:17	well # 2		19.2°		✓				
3	" "	9:21	well # 3		19.6°		✓				
4	" "	9:25	well # 4		19.0°		✓				
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	George Avila	Avila Family Dairy	6-7-23 9:57			
2nd		DENELE ANALYTICAL, INC.	6-7-23 10:00			T3158045

Required to be submitted to any agency? (YES / NO)

2023 Annual Report AP Avila

Final Audit Report 2024-05-05

Created:	2024-05-04
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAADXeVa21ppJ-r-Jm3gIdXSoAc1zqy4hoh

"2023 Annual Report AP Avila" History

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